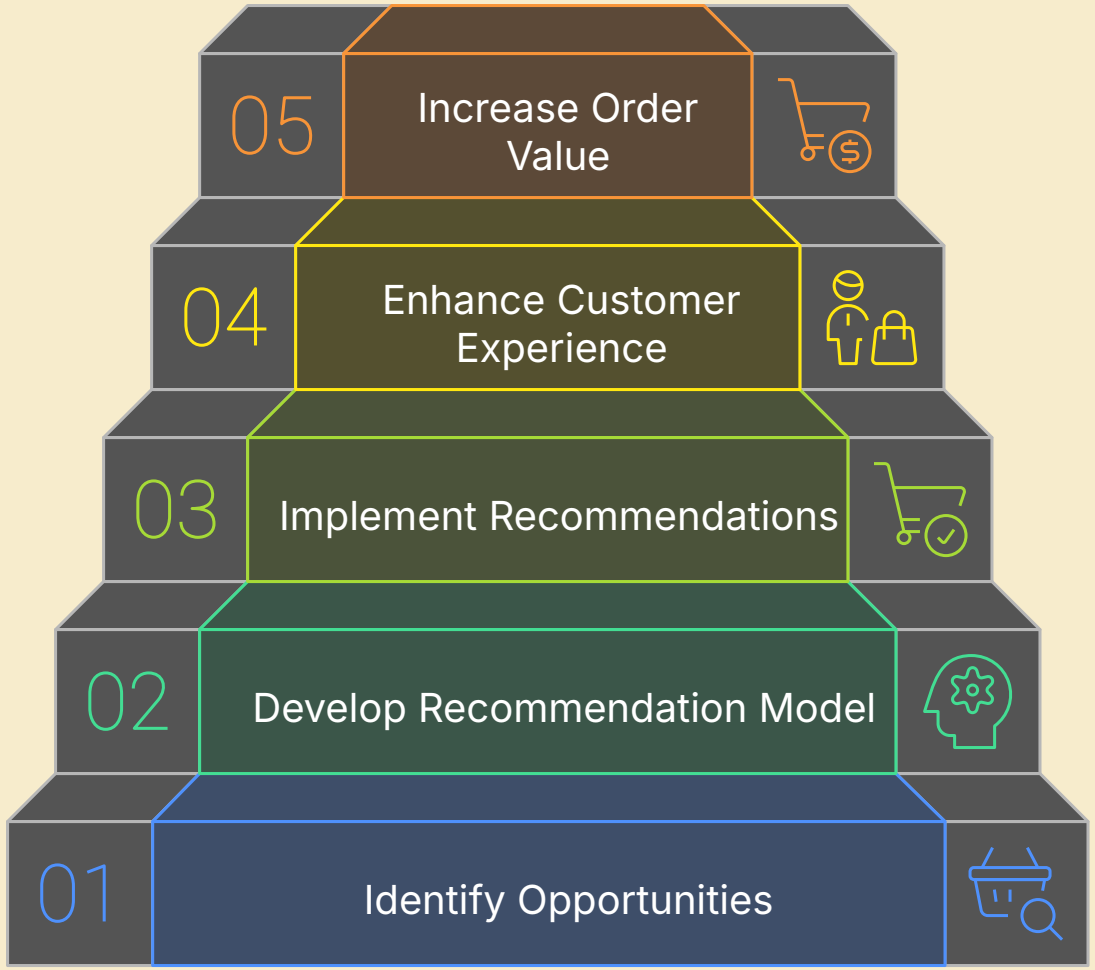
World Wide Technology

WWT Unravel 2025

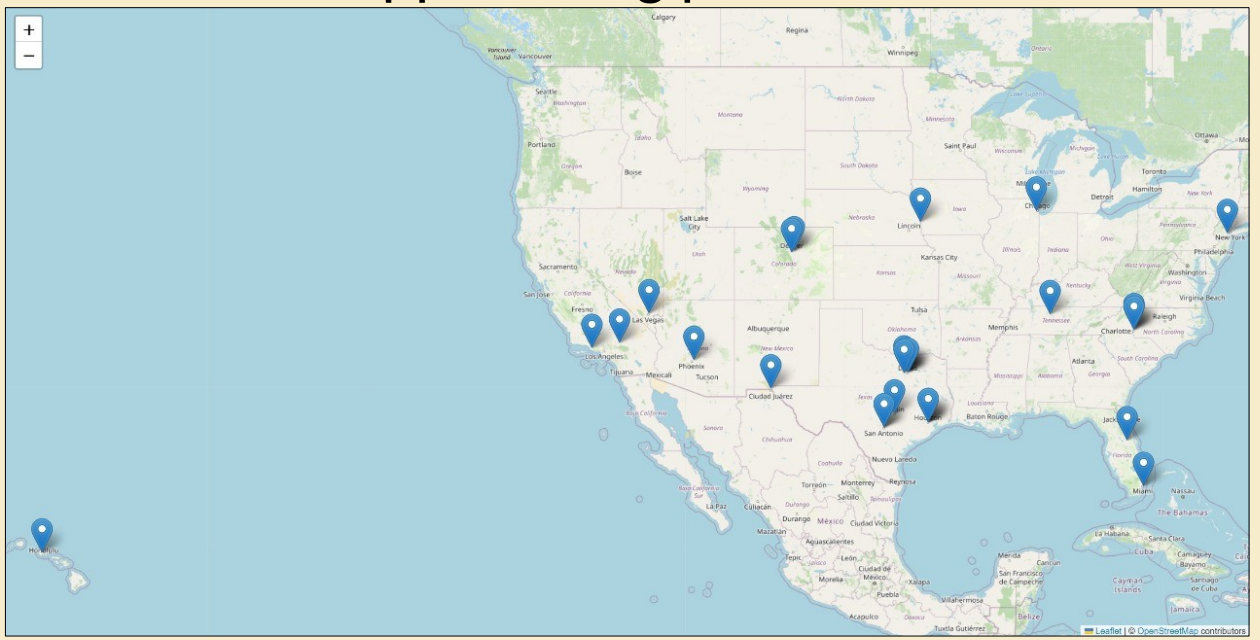
WINGS R US



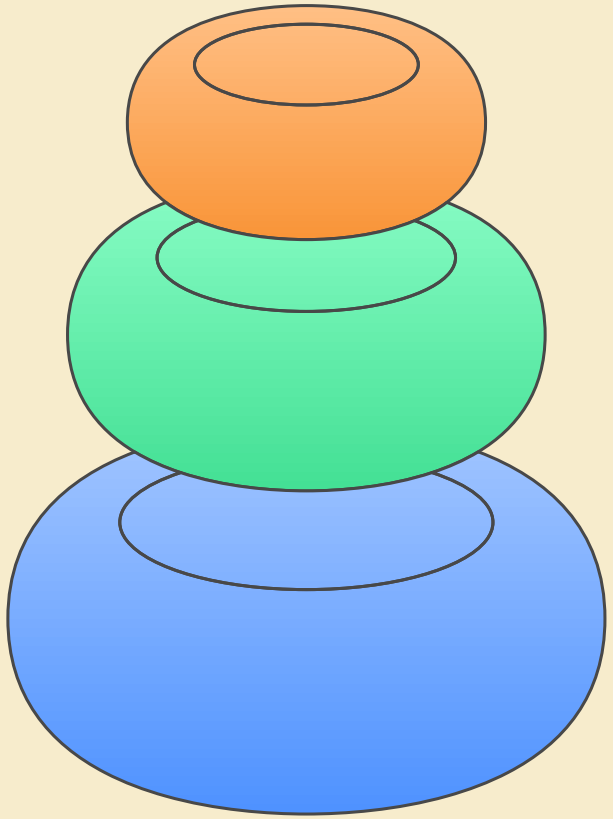
Business Situation



Store locations across the USA, mapped using postal codes



Dataset



Order Data

Details of customer orders

Customer Data

Information about customers

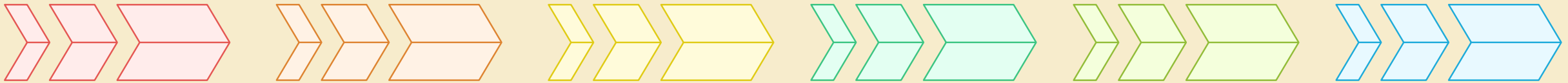
Store Data

Location-related information about stores

Best selling 3-item combinations

Item 1	Item 2	Item 3
Grilled Wings (All)	Ranch Dip (All sizes)	Spicy Wings (All)
Grilled Wings (All)	Ranch Dip (All sizes)	Spicy Feast Deal (All)
Grilled Wings (All)	Ranch Dip (All sizes)	Crispy Strips (All)
Grilled Wings (All)	Ranch Dip (All sizes)	Buffalo Fries (All)
Grilled Wings (All)	Spicy Wings (All)	Spicy Feast Deal (All)
Grilled Wings (All)	Spicy Wings (All)	Crispy Strips (All)

Pre-processing



Identify Non-Order Items

Recognizing items that are not actual order items

Remove Non-Order Items

Eliminating items that do not represent orders

Identify Rows with No Customer Type

Finding rows where customer type is missing

Calculate Average Order Price

Determining the average price of orders

Calculate Average Price per Item

Determining the average price per item

Classify Rows Using Euclidean Distance

Using Euclidean distance to classify rows

Modelling

5

Evaluate Model

Assess model performance using Recall@3 on a test set.



4

Validate Suggestions

Test suggestions against a sample order to ensure accuracy.



3



Predict Next Items

Suggest up to 3 likely next items based on input.

2



Association Rules

Generate rules from frequent item-sets, filtered by confidence.

1



Frequent Item-sets

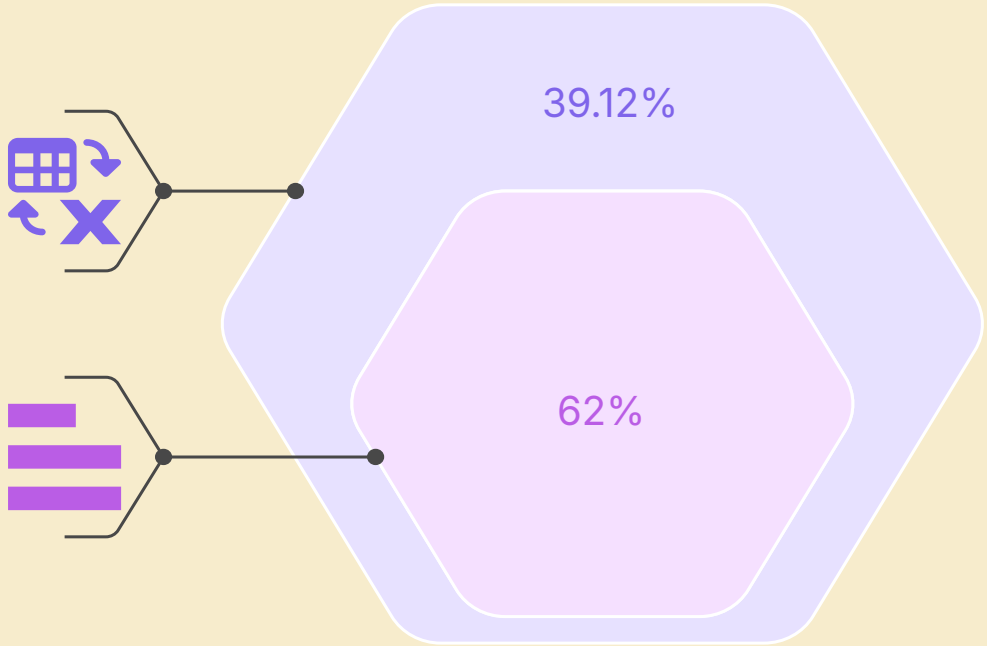
Identify common item combinations using FP-Growth.

Metrics

Recall@3 in predicting next item

Initial Dataset

Initial recall reflects the model's ability to correctly predict at least one of the next three items based on three given items in a large item space.



Refined Dataset

When the dataset was reduced by removing size variations and limiting item names to 48 distinct items, the recall improved

Conclusion

Key Factors in Recommendation System Success



Data Preprocessing

Crucial for improving recommendation accuracy and recall.

FP-Growth and Association Rules

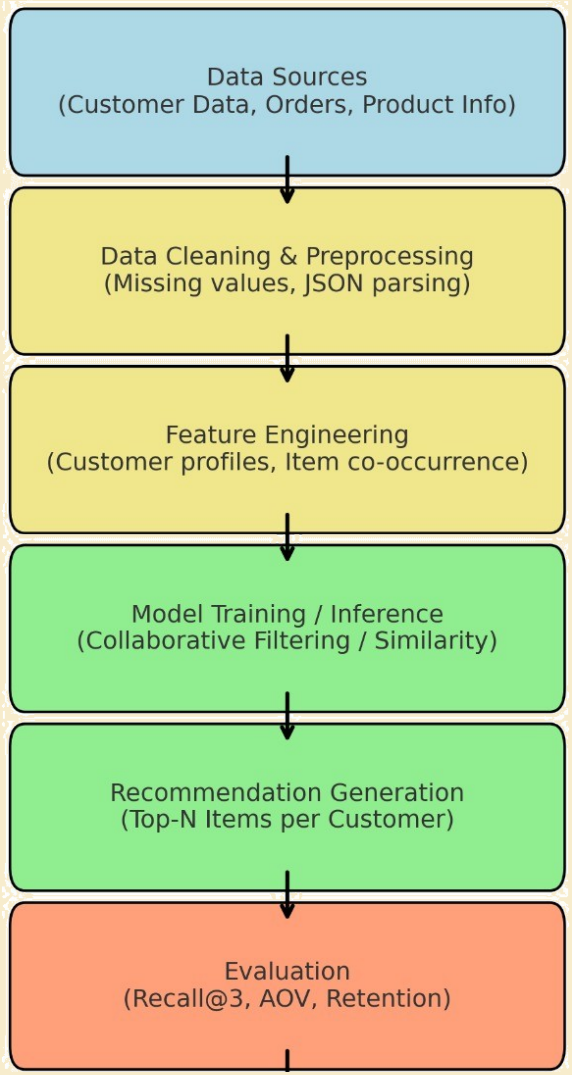
Captures frequent purchasing patterns effectively.

Model Performance

Demonstrates solid performance in real-world retail scenarios.

Appendix

High-Level Diagram



Github Repository:
https://github.com/PRANSH18/case-competitions

Limitations:

- Model performance may degrade for cold start customers (new shoppers).
- Real-time inventory changes are not yet integrated.
- Recommendations are limited to items present in the training data.

Best selling items at Wings R Us

Category	Orders
Grilled Wings (All)	545468
Ranch Dip (All sizes)	281835
Spicy Wings (All)	259766
Spicy Feast Deal (All)	199089
Crispy Strips (All)	170982
Buffalo Fries (All)	120897
Fried Corn - Regular	43385
Chicken Sub Combo	42434
Flavor Platter	41519
Chicken Sub	33425